BHUSHAN PATANKAR

CAD Developer (Sr.Engineer)

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PROFESSIONAL SUMMARY

7+ Years Experienced Technical professional dedicated to driving engineering efficiency by bridging software development with mechanical design. Specializes in building custom CAD automation tools and programmatic workflows utilizing the SolidWorks API, C#, and DriveWorks. Passionate about transforming traditional engineering cycles into highly optimized systems through robust Win-Forms applications.

PORTFOLIO LINK

WORK EXPERIENCE

Tech Mahindra | 07-2025 to Present

Sr. Engineer (CAD Developer / CAD Automation)

- **Advanced CAD Automation**: Developed robust, scalable configurator solutions using DriveWorks Pro, seamlessly capturing and controlling complex SolidWorks assemblies
- **Steered an 8-person team** acting as the unofficial lead–to ensure the **on-time delivery** of projects
- Actively conducted **Technical meeting** and **Project Progress** Meeting with teams and Stakeholders.
- Conceptualized and implemented innovative workflow solutions, reducing repetitive drafting tasks by **80%**.
- **Data Management**: Integrated DriveWorks with external databases and spreadsheets to dynamically pull material properties, pricing, and sizing constraints into the configurator.
- **Macro & Script Execution**: Utilized DriveWorks to trigger custom SolidWorks macros and API routines, extending out-of-the-box functionality for highly specific engineering tasks.
- **Decreased execution time** by optimizing models, removing global variables, and applying de-featuring techniques

Forms and Surfaces India Pvt Ltd | 01-2022 to 06-2025

Design and Automation Engineer

- Support Design team to minimize repetitive Tasks using **Solidworks macro**.
- Created **DXF Creator** which eliminates manual DXF export and QC team effort.
- Maintain design Documentation and standardize existing process for **future design automation scope**.
- Created **Master Model** of various products in Solidworks to use as reference for new orders.
- Created standard **part libraries** to use as it is in new products

Prime Technologies | 12-2020 to 01-2022

Design Engineer and Project Coordinator

- **Domain**-Material Handling, Process Automation, SPM, Industrial Lift, Heavy Structures

Cognition Technologies | 07-2019 to 12-2020

Design Engineer and Project Coordinator

- **Domain**-Material Handling, Conveyor

SKILLS

- **CAD Automation & Development**: SolidWorks API, DriveWorks (Advanced, CPQ, Autopilot, Data Management), C#, Win-Forms, Master Modeling
- **Software & IT**: C#, .NET Core, MVC, OOP, Python, VBA, SQL, Visual Studio, AI Fundamentals
- **Mechanical Engineering**: Product Design, DFM/DFA, Static Structural Analysis, Reverse Engineering, Drafting
- **CAD & CAE Tools**: SolidWorks (Expert), AutoCAD (Intermediate), CATIA (Basic), UG-NX (Basic), ANSYS (Basic)
- **Industrial Operations**: Sheet Metal Fabrication, Material Handling Automation, DFMEA, BOM-Routing, Design Calculations, Job Orders, Technical Documentation
- **Professional Skills**: Team Leadership, Complex Problem Solving, Logical Architecture, Creative Thinking, Technical Presentation

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PROJECTS

GA Modernization (Client: Flowserve US, Via Tech Mahindra) | 02-2026 to present

- Automated the generation of General Arrangement (GA) drawings for **Pump Motor Skids** and auxiliary components (seals, gearboxes, junction boxes) using DriveWorks and SolidWorks.
- **Modernized a legacy automation tool** by migrating it to DriveWorks, introducing an updated user interface and modular architecture.
- Achieved a **75% reduction in generation time** through workflow optimization, CAD model de-featuring, and global variable elimination.
- Accelerated development and testing cycles by implementing strict **modular logic separation**.
- Restructured back-end **databases** and optimized stored procedures to **improve system performance**.
- **Integrated proprietary Excel-based engineering calculations** directly into the DriveWorks rules engine.
- **Extended native DriveWorks capabilities** by developing custom macros for automated dimension arrangement

Valve Generator (Client: Flowserve US, Via Tech Mahindra) | 11-2025 to 02-0206

- Engineered a **web-based application** to automatically generate 3D CAD models and catalog drawings for customers and sales teams.
- Architected **multi-child workflows** incorporating a repository of over **1,000 master models** of valves and actuators.
- Executed **rigorous testing** across 200+ configuration combinations to ensure output quality, verifying proper **dimensional accuracy and required drawing views**.
- **Deployed** the final automation tool into a live production environment running on an **Autopilot server**

GAD Tool (Client: Flowserve US, Via Tech Mahindra) | 07-2025 to 10-2025

- Developed an automation tool to dynamically generate 3D CAD models and General Arrangement (GA) drawings for Vertical Pumps based on custom input specifications.
- I**ntegrated cross-program data workflows**, enabling the system to automatically ingest Excel values populated by external applications to output required engineering files.
- Architected the foundational master CAD models and programmed robust logical rule sets to drive accurate, automated model generation.
- **Configured Autopilot server environments** and successfully deployed the finalized tool into production strictly within the scheduled project timeline

DXF Creator Tool (Forms and surfaces)

- Developed a **C# Win-Forms application** leveraging the **SolidWorks API** to automate the extraction of customized DXF files for laser cutting operations.
- Engineered batch-processing functionality to handle multiple parts simultaneously, accurately extracting flat pattern views while maintaining exact orientation.
- Automated the generation of clean, production-ready DXF outputs directly to user-specified directories.
- Achieved a **90% reduction in processing time** and completely **eliminated manual errors** –such as incorrect configurations and orientation mistakes thereby reducing downstream production rework

Drawing Creator (Forms and surfaces)

- Developed a C# Win-Forms application to **fully automate the generation of drawings**, PDFs, and DXFs for complex control panel and bus bar assemblies.
- Engineered an automated workflow that recursively extracts all constituent parts from a user-selected assembly to generate comprehensive drawings with **standard views, BOMs, and dimension arrangements**.
- **Programmed custom geometric algorithms** to automatically detect part edges and intelligently apply accurate dimensions to scratch-built drawings.
- Leveraged CAD custom properties to **dynamically classify parts** and trigger specific, rule-based drawing generation methods.

Macro Library for Design Engineers (Forms and surfaces)

- **Drawing Template Batch Replace** during rebranding of company with new logo and layout.
- **Bend Parameter**: Single click bend parameter update based on excel integration with SolidWorks

Product Design

- Sheet metal wall cladding systems, Quartz cladding for Apple Store, Lift interiors cladding, Outdoor products like benches, light Bollard, Light planes
- Material Handling -25 Tons battery operated RGV, Hydraulic Lift, Lift structure, Scissor Lift, Turn Table, Conveyor system, Pipe handling system.
- All Terrain Vehicle for BAJA2019 event

KEY ACHIEVEMENTS

- Got **Pat on the back award** at Tech Mahindra (2025)
- **Employee of the month** Award Forms and Surfaces(2024)
- **Rookie of the year 2022** award by Forms and Surfaces. (2023)
- **Special Recognition Award** (monthly) by Forms and Surfaces (2022)
- Active participator as **Design team lead** of ATV for BAJA2019 event held at IIT , Ropar (2019)

TRAINING /COURSES

- **CDWP** Certified Driveworks Professional
- **Solidworks API** with C# WInforms by FeesWorth
- Certified SOLIDWORKS Professional **CSWP** , Certified SOLIDWORKS Professional in Advance sheet metal **CSWPA-SM** by Dassult System (2020)
- Internship at Tragen Pvt. Ltd. Aurangabad (2017)
- SolidWorks training IGTR,NAGPUR (2018)
- Attended workshop on **5-S** and **Kaizen** At Endress+Houser at Aurangabad (2018)

EDUCATION

Bachlor of Mechanical Engineering | 2015-2019

Shri Sant Gajanan Maharaj College of Engineering, Shegaon, MH

HOBBIES

- Sketching Portraits
- Book Reading
- Swimming
- Riding

LANGUAGES

- English
- Hindi
- Marathi

DECLEARATION

I hereby declare that above information is true as per my best knowledge and belief

Bhushan Patankar